

## Lab. 3

### *Comparators and the Schmitt Trigger*

Team No. 2

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#### Reference:

Floyd Thomas L., Buchla David M.: *Experiments in Analog Fundamentals, A System Approach*. Pearson Education, 2013.

#### Reading:

Floyd and Buchla, *Analog Fundamentals: A Systems Approach*, Section 8-1.

#### Objectives:

After performing this experiment, you will be able to:

1. Compare the input and output waveforms for comparator and Schmitt Trigger circuits.
2. Use the oscilloscope to plot the transfer curve for a comparator circuit, including one with hysteresis.
3. Construct and test relaxation oscillator using a Schmitt trigger.

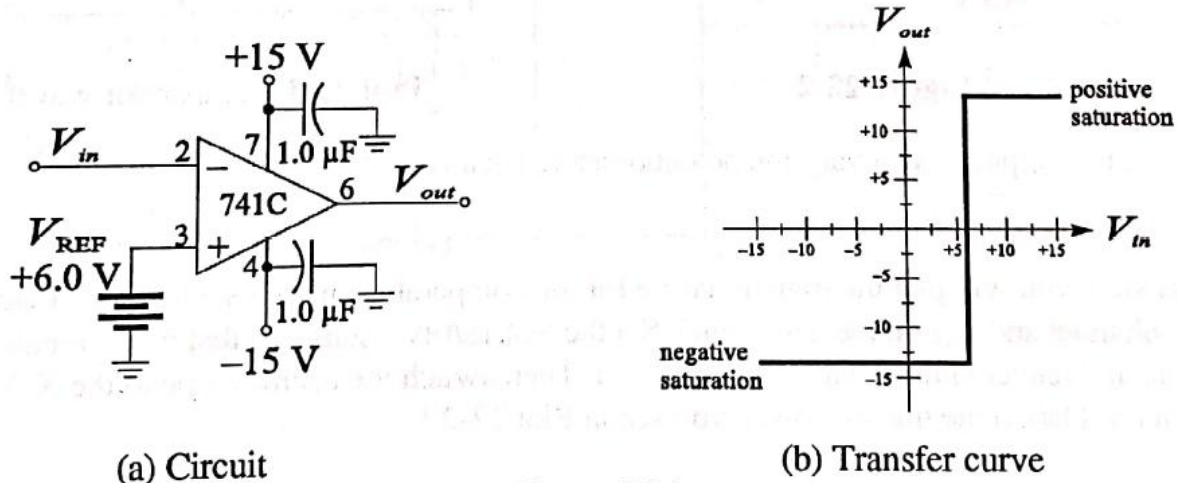
#### Materials Needed:

- Resistors: one 100 k $\Omega$
- Two 1.0  $\mu$ F capacitors.
- One 10 k $\Omega$  potentiometer.
- One 741C op-amp
- *For Further investigation:*
  - One additional 100 k $\Omega$  resistor.
  - One 0.1  $\mu$ F capacitor.

## Summary of Theory

A comparator is a switching device that produces a high or low output, depending on which of the two inputs is larger. The comparator is run with the very high open-loop gain. When the noninverting input is very slightly larger than the inverting input, the output goes to positive saturation; otherwise, it goes to negative saturation. Although general-purpose op-amps can be used as comparators, specially designed op-amps can switch faster and have additional features not found on general-purpose op-amps. For noncritical applications, general-purpose op-amps are satisfactory and will be used in this experiment.

A comparator circuit is characterized by its transfer characteristic. The transfer characteristic (curve) is a plot of the output voltage (plotted along the  $y$ -axis) as a function of the input voltage (plotted along the  $x$ -axis). Consider the comparator shown in Figure 1. The reference voltage is  $+6\text{ V}$ . When the input is greater than  $+6.0\text{ V}$ , the output will go to positive saturation (approximately  $+13\text{ V}$ ); when the input is less than  $+6.0\text{ V}$ , the output will be in negative saturation. A glance at the transfer curve will show the output for any given input voltage.



**Figure 22-1**

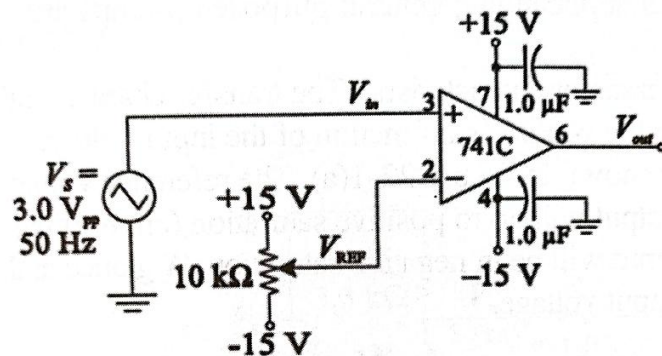
*Figure 1: Comparator circuit (a) and its transfer curve (b)*

Because of the sensitivity to a small input change, the output of a comparator may change due to noise on the input or when the input changes slowly. To avoid this, hysteresis is added to the comparator circuit by introducing positive feedback. The circuit, called a *Schmitt trigger*, has two switching thresholds – one for a rising input voltage, the other for a falling input. By separating the two thresholds, noise effects can be eliminated. In this experiment, you will investigate two comparators and an inverting Schmitt trigger circuit.

## Procedure

### Comparator and Comparator Transfer Curve

- Figure 2 shows an inverting comparator circuit with a variable threshold determined by the potentiometer setting. Construct the circuit and set  $V_{REF}$  to near 0 V. Set the function generator for a  $3V_{pp}$  triangle waveform at 50 Hz and observe the input and output waveforms on a two-channel oscilloscope. Sketch the waveforms on Plot 22-1. Note the point where switching takes place. Be sure to label all plots in this experiment.



**Figure 22-2**

*Figure 2: Inverting comparator with variable threshold*



*Figure 3: Plot 22-1 — Comparator waveform.*

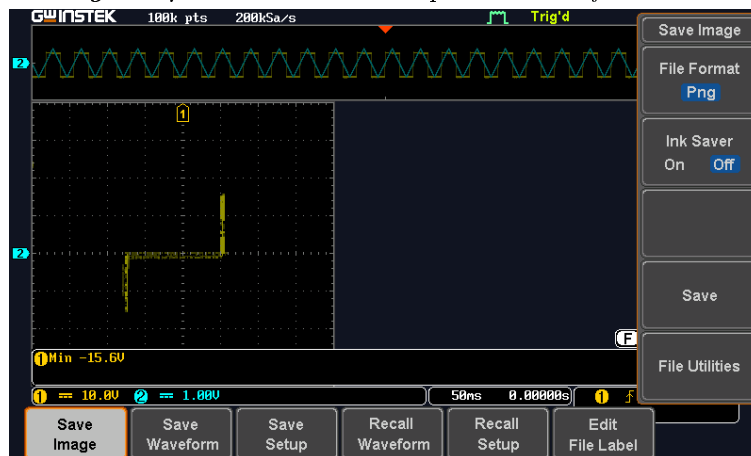
- Observe the output as you vary the potentiometer. Then reset  $V_{REF}$  to 0 V.

Observations: Se observa una señal cuadrada que cambia su valor (de alto a bajo) y el ancho de pulso aumenta al variar el potenciómetro.

3. In this step, you will plot the transfer curve for the comparator on the oscilloscope. Place  $V_{in}$  on the X-channel and  $V_{out}$  on the Y-channel. Set the *VOLTS/DIV* control so that both signals are on the screen. Neither channel should be inverted. Then switch the oscilloscope to the X-Y mode. Sketch (and label) the transfer curve you see in Plot 22-2.



*Figure 4: Plot 22-2 — Comparator transfer curve.*



*Figure 5: Plot 22-3 — Comparator transfer curve (inputs reversed).*

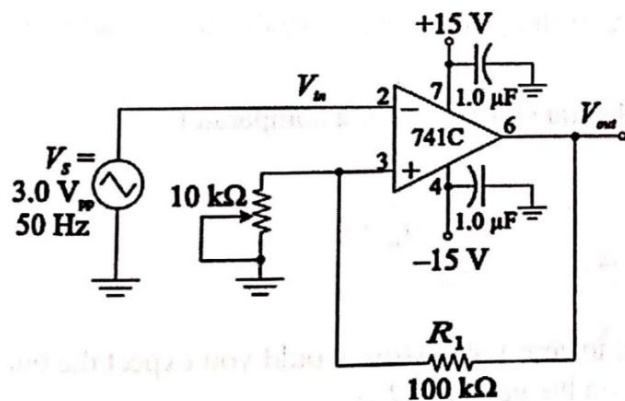
4. Vary the potentiometer as you observe the transfer curve.

Observations: Se observa como la curva de transferencia se satura positivamente al variar el potenciómetro. En la señal de la curva inversa se observa como se satura al aumentar el voltaje de referencia.

5. While observing the transfer curve, reverse the inputs to the comparator. Sketch the new transfer curve in Plot 22-3.

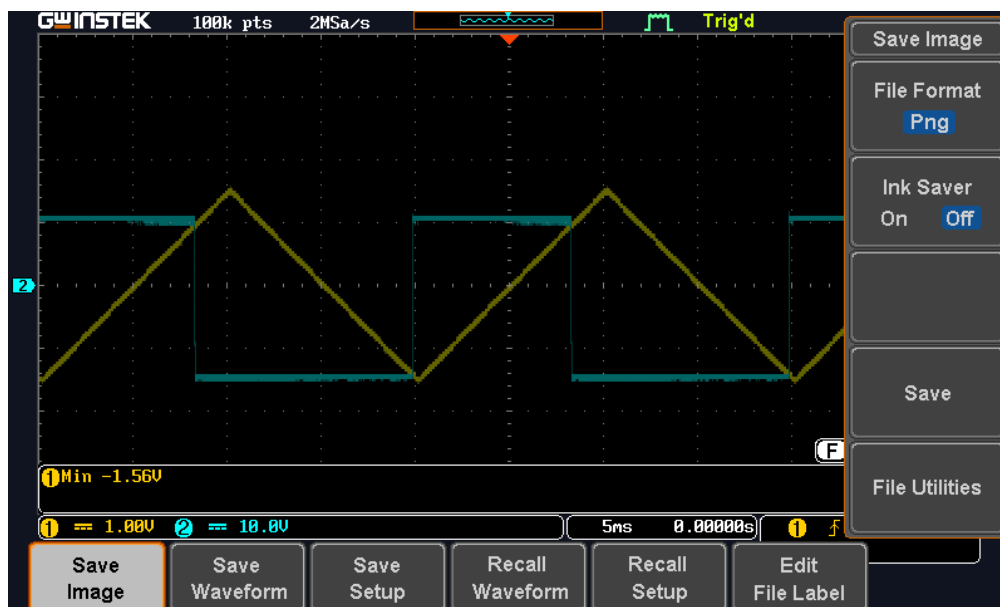
### Schmitt Trigger and Schmitt Trigger Transfer curve

- Construct the Schmitt trigger circuit shown in Figure 6. Set the potentiometer to the maximum resistance and put the oscilloscope in normal time base mode (not X-Y mode). Slowly reduce the resistance of the potentiometer and observe the input and output waveforms. Note that when the output changes states, the input voltage is different for a rising and a falling signal. Sketch the observed input and output waveforms in Plot 22-4.



**Figure 22-3**

*Figure 6: Inverting Schmitt trigger circuit*



*Figure 7: Plot 22-4 — Schmitt trigger waveform.*

7. In this step, you will plot the transfer curve for the Schmitt trigger on the oscilloscope. The input signal is again on the X-channel and the output signal is on the Y-channel. Select the X-Y mode and adjust the controls to view the transfer curve. Notice the hysteresis. Sketch (and label) the transfer curve you see in Plot 22-5.
8. While observing the transfer curve, vary the potentiometer.

Observations: En la curva de transferencia de este circuito se observa una "caja", ya que el circuito posee histéresis, ya que es un circuito retroalimentado.



Figure 8: Plot 22-5 — Schmitt trigger transfer curve.

## Conclusion

## Evaluation and Review Questions

1. Describe how the threshold voltage changes the transfer curve for a comparator.
2. Assume the circuit in Figure 2 had  $V_{REF}$  set to zero volts. How would you expect the output to be affected by varying the dc offset control on the generator?
3. Would a sinusoidal input to the comparators produce the same transfer curve as a triangle waveform? Explain.
4. Summarize the important differences between a comparator and Schmitt trigger.
5. Assume the input signal in Figure 6 could have as much as  $100\text{ mV}_{pp}$  noise. To avoid multiple tripping due to noise, you need to set the trip points at least  $100\text{ mV}$  apart. What is the minimum value of resistance that the potentiometer can be set? Assume the output saturates at  $\pm 13\text{ V}$ .

## For Further Investigation

The Schmitt trigger you investigated in the experiment can be modified slightly to form a relaxation oscillator. Investigate the relaxation oscillator shown in Figure 9. (For simplicity, power supply connections are not shown.) Look at the waveform across the capacitor and at the output. Test the effect of the potentiometer on the frequency and waveform on the capacitor. Normally, a triangle waveform is observed. Under what conditions can you obtain a sine wave? Then observe the transfer curve by observing the signal on pin 2 and pin 6 in the X-Y mode, can you explain the shape? Summarize your findings in a short report.

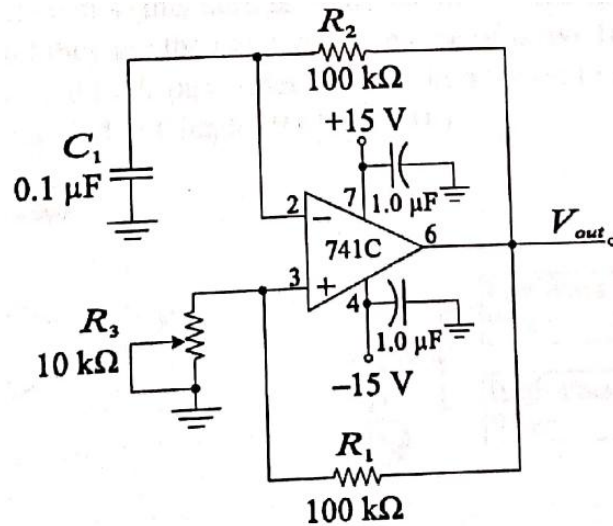
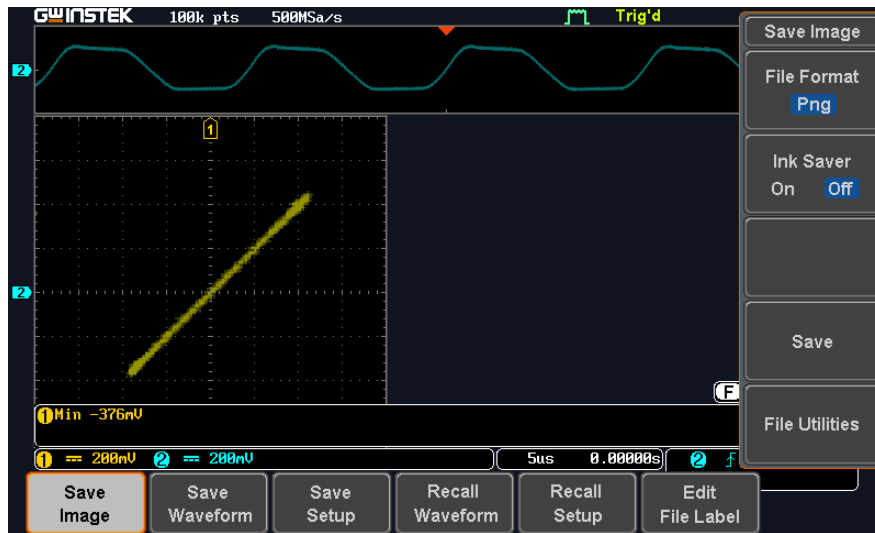


Figure 22-4

Figure 9: Relaxation oscillator based on the Schmitt trigger



Figure 10: Plot 22-6 — Schmitt trigger waveform as an oscillator.



*Figure 11: Plot 22-6 — Schmitt trigger transfer curve as an oscillator.*

Observatio: Se puede obtener una onda senoidal cuando se ajusta la resistencia a  $0\omega$  o muy baja, con eso el slew rate se lleva al límite y no se pueden generar los cambios bruscos de una onda cuadrada y con eso se convierte en un filtro pasa-bajas. Por otro lado, en la curva de transferencia se observa una línea diagonal ya que el voltaje de entrada y el de referencia son exactamente iguales